

Assignment 2

Diabetes Prediction using Machine Learning

Part 1 — Data Preparation

1.1 Dataset Exploration Summary

The dataset was loaded using the Pandas library and explored to understand its structure and characteristics.

Dataset Shape

The dataset contains:

- **768 rows**
- **9 columns**

Therefore, there are 768 patient records available for the analysis.

Column Names

The dataset contains the following columns:

1. Pregnancies
2. Glucose
3. BloodPressure
4. SkinThickness
5. Insulin
6. BMI
7. DiabetesPedigreeFunction
8. Age
9. Outcome

Data Types

Most variables in the dataset are numerical.

- Integer (`int64`) variables include Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, Age, and Outcome.
- Floating-point (`float64`) variables include BMI and DiabetesPedigreeFunction.

Basic Statistical Summary

A statistical summary was generated using the `describe()` function. Some important observations include:

- The average Glucose value is approximately **120.89**.
- The average BMI is approximately **31.99**.
- The average age of patients is approximately **33.24 years**.
- The Outcome mean is approximately **0.349**, indicating that around 35% of the records belong to the diabetes-positive class.

1.2 Features and Target Variable

The dataset was divided into input features and the target variable.

Features (X)

The following eight columns were used as input features:

- Pregnancies
- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI
- DiabetesPedigreeFunction
- Age

Target Variable (y)

The **Outcome** column was selected as the target variable.

- `0` = No Diabetes
- `1` = Diabetes

1.3 Preprocessing Steps Performed

The following preprocessing activities were considered and performed:

- The dataset was checked for missing values.

- No formally missing values were found.
- All features were numerical, so categorical encoding was not required.
- The data was separated into features and target variables.
- The dataset was divided into training and testing datasets.

1.4 Training and Testing Dataset Sizes

After performing the train-test split:

- **Training dataset:** 652 records with 8 features
 - **Testing dataset:** 116 records with 8 features
-

Part 2 — Machine Learning Model Building

2.1 Models Implemented

Three Machine Learning classification algorithms were implemented:

1. **Logistic Regression**
2. **Decision Tree Classifier**
3. **Random Forest Classifier**

All models were trained using the prepared training dataset.

2.2 Logistic Regression

Logistic Regression is suitable for binary classification problems because the target variable has two possible classes: diabetes and no diabetes.

The model estimates the probability that a patient belongs to the diabetes-positive class based on the input medical features.

2.3 Decision Tree Classifier

The Decision Tree Classifier makes predictions by splitting the data into branches based on feature values. It learns decision rules from the training data and classifies patients into either the diabetes or non-diabetes category.

A Decision Tree can capture non-linear relationships between the input features and the target variable.

2.4 Random Forest Classifier

The Random Forest Classifier is an ensemble learning algorithm that combines predictions from multiple decision trees.

Each tree makes a prediction, and the final prediction is determined using the combined results of the trees. Random Forest is suitable for this problem because it can model complex relationships and generally provides more robust predictions than a single decision tree.

2.5 Model Performance Comparison

The performance of the three models on the testing dataset was compared.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.818966	0.738095	0.756098	0.746988
Decision Tree	0.801724	0.687500	0.804878	0.741753
Random Forest	0.827586	0.733333	0.804878	0.767442

Based on the comparison, **Random Forest was selected as the best-performing model** because it achieved the highest accuracy and F1-score among the implemented models.

Part 3 — Model Training

3.1 Model Training

The selected Random Forest model was trained using the prepared training dataset.

The model was successfully fitted using the training features and target values.

3.2 Training Predictions

Predictions were generated using the training dataset to evaluate how well the Random Forest model learned the patterns present in the training data.

3.3 Training Performance

The training performance of the Random Forest model was as follows:

Metric	Training Performance
Accuracy	0.967791
Precision	0.985849
Recall	0.920705
F1-Score	0.952164

The training accuracy is approximately **96.78%**, which indicates that the model performs very well on the training data.

3.4 Overfitting and Underfitting Analysis

The training performance was compared with the testing performance.

Metric	Training	Testing
Accuracy	0.967791	0.827586
Precision	0.985849	0.733333
Recall	0.920705	0.804878
F1-Score	0.952164	0.767442

The model performs significantly better on the training dataset than on the testing dataset. Therefore, the results indicate **potential overfitting**.

This means that the model has learned the training data very effectively but its performance decreases when predicting previously unseen test data.

Part 4 — Model Evaluation

4.1 Test Dataset Performance

The selected Random Forest model was evaluated using the testing dataset.

The following results were obtained:

Metric	Test Performance
Accuracy	0.827586
Precision	0.733333
Recall	0.804878
F1-Score	0.767442

The model achieved a test accuracy of approximately **82.76%**.

4.2 Interpretation of Performance Metrics

Accuracy

Accuracy represents the overall proportion of correct predictions.

The model achieved an accuracy of approximately **82.76%**, showing reasonably good overall predictive performance.

Precision

The precision value is approximately **73.33%**.

This indicates how many patients predicted as diabetes-positive were actually diabetes-positive.

Recall

The recall value is approximately **80.49%**.

Recall is particularly important in a diabetes prediction problem because it measures the model's ability to correctly identify patients who actually have diabetes.

F1-Score

The F1-score is approximately **76.74%**.

This metric provides a balance between Precision and Recall.

4.3 Confusion Matrix

The confusion matrix obtained from the test dataset is:

Actual / Predicted	No Diabetes	Diabetes
No Diabetes	63	12
Diabetes	8	33

Therefore:

- True Negatives (TN): 63
- False Positives (FP): 12
- False Negatives (FN): 8
- True Positives (TP): 33

4.4 Confusion Matrix Interpretation

The model correctly identified:

- 63 patients without diabetes.
- 33 patients with diabetes.

However:

- 12 patients without diabetes were incorrectly predicted as having diabetes.
- 8 patients with diabetes were incorrectly predicted as not having diabetes.

The False Negative value is particularly important because a patient with diabetes may be incorrectly classified as not having diabetes.

Part 5 — Model Performance Analysis

5.1 Most Important Performance Metric

For a diabetes prediction problem, **Recall** is considered one of the most important performance metrics.

Recall measures the ability of the model to correctly identify patients who actually have diabetes.

A high Recall is important because failing to identify a patient with diabetes can be more serious than incorrectly identifying a patient as potentially having diabetes. Such patients may require further medical testing, whereas a missed diabetes case could delay further diagnosis.

The Random Forest model achieved a Recall of approximately **80.49%**, which means that it successfully identified a large proportion of the actual diabetes-positive cases.

5.2 Overall Model Performance

The Random Forest model performed reasonably well on the testing dataset.

Key test results are:

- Accuracy: **82.76%**
- Precision: **73.33%**
- Recall: **80.49%**
- F1-Score: **76.74%**

Compared with Logistic Regression and Decision Tree, Random Forest achieved the best overall combination of Accuracy and F1-Score.

However, the difference between training accuracy (**96.78%**) and testing accuracy (**82.76%**) suggests potential overfitting.

5.3 Key Findings

The main findings of this project are:

1. The dataset contains 768 records and 8 input features with Outcome as the target variable.
2. No formally missing values were found in the dataset.
3. Three Machine Learning models were implemented and compared.
4. Random Forest achieved the best overall performance among the three models.
5. The Random Forest model achieved approximately 82.76% test accuracy.
6. The model achieved a Recall of approximately 80.49%, indicating good ability to identify diabetes-positive patients.
7. The training performance was significantly higher than the testing performance, suggesting potential overfitting.

5.4 Model Limitations

The following limitations were identified:

1. Potential Overfitting

The training accuracy is approximately 96.78%, while the testing accuracy is approximately 82.76%. This difference suggests that the model may have learned the training data too closely.

2. Limited Dataset Size

The dataset contains only 768 records. A larger and more diverse dataset may improve the ability of the model to generalize to new patients.

3. False Negative Predictions

The model produced 8 False Negative predictions. In a diabetes prediction application, incorrectly predicting that a patient does not have diabetes could be an important limitation.

4. Scope of Available Features

The prediction is based only on the available medical features in the dataset. Additional relevant patient information may improve prediction performance.

5.5 Recommendations for Improvement

The following approaches can be used to improve the model:

1. Hyperparameter Tuning

The Random Forest model parameters can be optimized using techniques such as GridSearchCV or RandomizedSearchCV. Parameters such as the number of trees, maximum tree depth, and minimum samples per split can be tuned.

2. Feature Engineering and Feature Selection

New meaningful features can be created, and less useful features can be removed. This may reduce noise and improve model generalization.

3. Cross-Validation

Cross-validation can be used to evaluate the model on multiple splits of the dataset and obtain a more reliable estimate of model performance.

4. Handling Class Imbalance

The class distribution can be further analyzed. Techniques such as class weighting or resampling can be considered if class imbalance affects the model's ability to identify diabetes-positive patients.

5. Try Other Classification Algorithms

Other algorithms, such as Support Vector Machines, Gradient Boosting, or XGBoost, could be evaluated and compared with the current models.